Supplementary Information

Qualification of Pathology Foundation-Model Signals Across Cohorts, Sites and Clinical Endpoints

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Supplementary Methods and Results

This document reports the saved extended analyses that support, but do not enlarge, the claims in the main manuscript. Patient-level uncertainty and repeated tile-sampling variation are kept distinct. Missing or unevaluable outcomes were never treated as negative labels, and all endpoint names retain their source definitions.

Abbreviations. CONCH denotes CONtrastive learning from Captions for Histopathology; NADT denotes neoadjuvant androgen-deprivation therapy; PANDA denotes Prostate cANcer graDe Assessment; TCGA-PRAD denotes The Cancer Genome Atlas prostate adenocarcinoma cohort; and TCGA-CDR denotes the TCGA Pan-Cancer Clinical Data Resource. Area under the receiver operating characteristic curve (AUROC), time-dependent area under the curve (td-AUC), concordance index (C-index), integrated Brier score (IBS), progression-free interval (PFI), progression-free survival (PFS), disease-free survival (DFS), hematoxylin and eosin (H&E), copy-number alteration (CNA), confidence interval (CI), and 256-bit Secure Hash Algorithm (SHA256) are used below. Gene/protein symbols are expanded as follows: phosphatase and tensin homolog (PTEN), speckle type BTB/POZ protein (SPOP), androgen receptor (AR), tumor protein p53 (TP53), RB transcriptional corepressor 1 (RB1), serine peptidase inhibitor Kazal type 1 (SPINK1), ETS variant transcription factors 1 and 4 (ETV1 and ETV4), and ETS transcription factor ERG (ERG). The post-hoc Cox recurrence-risk score called “marker 7” in legacy artifacts is termed the post-hoc recurrence risk signal here. CH, EJ, G9, HC, KK, and YL are deidentified source institution codes rather than expanded institution names.

Supplementary Table S1 records the complete audited endpoint hierarchy without renaming reconstructed outcomes as official PFI.

Table S1: Endpoint hierarchy retained from the audited endpoint source.

ID	Hierarchy	Endpoint	Primary metric
E01	primary	continuous marker qualification	patient Spearman rho
E02	primary	binary marker qualification	patient AUROC
E03	secondary	PTEN/AR held-out increment	delta AUROC or delta R2
E04	exploratory	reconstructed with-tumor endpoint	C-index; td-AUC; IBS; calibration
E05	exploratory	strict recurrence-only endpoint	C-index; td-AUC
E06	exploratory	TCGA-CDR PFS	C-index; td-AUC; landmark AUC
E07	exploratory	TCGA-CDR DFS	C-index; td-AUC; landmark AUC
E08	exploratory	official TCGA-CDR PFI	C-index; endpoint concordance
E09	secondary	3-year and 5-year landmark	landmark C-index/AUROC

The complete 17-test family

This subsection enumerates the complete multiplicity family so that the main-text findings can be interpreted against all saved discovery-family tests rather than a selected subset.

The multiplicity family contained 13 marker hypotheses and four nested/refit confounder audits. Continuous marker hypotheses used two-sided Spearman tests, binary hypotheses used two-sided Mann–Whitney tests, and the refit-permutation audit used the saved one-sided test for positive improvement. Benjamini–Hochberg false-discovery-rate adjustment was applied jointly to these 17 rows. External or cross-encoder evaluations of an already counted hypothesis were qualification checks rather than additional discovery-family rows.

Table S2: Complete saved 17-row revision-analysis multiplicity family. Effects and probability values are copied from the audited source; q is the Benjamini–Hochberg value.

Test	Metric	Effect	p	q
① NADT H&E -> Gleason	Spearman rho	0.478	0.00207	0.00703
② NADT H&E -> Phenotype	Spearman rho	0.8	9.63×10^{-10}	1.64×10^{-8}
③ NADT ERG -> Gleason	Spearman rho	0.524	0.00061	0.00346
④ TCGA-PRAD H&E -> PTEN loss	AUROC	0.632	0.000562	0.00346
⑤ TCGA-PRAD H&E -> SPOP mutation	AUROC	0.519	0.742	0.788
⑥ TCGA-PRAD H&E -> AR score	Spearman rho	0.195	0.00122	0.0052
TCGA-PRAD H&E -> TP53 mutation	AUROC	0.485	0.835	0.835
TCGA-PRAD H&E -> TP53 CNA loss	AUROC	0.562	0.0981	0.208
TCGA-PRAD H&E -> RB1 CNA loss	AUROC	0.536	0.363	0.441
TCGA-PRAD H&E -> SPINK1 high	AUROC	0.618	0.152	0.288
TCGA-PRAD H&E -> ETV1 altered	AUROC	0.429	0.245	0.347
TCGA-PRAD H&E -> ETV4 altered	AUROC	0.687	0.023	0.0559
TCGA-PRAD H&E -> ERG fusion status	AUROC	0.526	0.458	0.519
PTEN / grade-only	AUROC change	0.0195	0.203	0.314
AR / grade-only	R-squared change	0.0044	0.176	0.299
Post-hoc recurrence risk / grade-only	C-index change	0.0619	0.01	0.0283
Post-hoc recurrence risk / fully adjusted	C-index change	0.00163	0.305	0.399

Supplementary Table S2 reports every row in the saved 17-row revision-analysis multiplicity family; no selected subset is presented as the family.

The family records the tests as originally defined. Later endpoint-matched, same-patient and same-bootstrap-draw comparisons are reported below and govern the narrower recurrence interpretation; they do not retrospectively replace rows in the saved multiplicity family.

Detailed discrimination summaries

We first expand the cross-cohort grade and phenotype evidence underlying the transportable state assigned in the main Results.

Supplementary Figure S1 shows the saved aggregate discrimination and association estimates for grade and phenotype transfer across NADT, PANDA and PRECISE. No receiver operating characteristic coordinates were stored for these rows, so curves were not reconstructed from aggregate values. Missing intervals remain visibly unavailable.

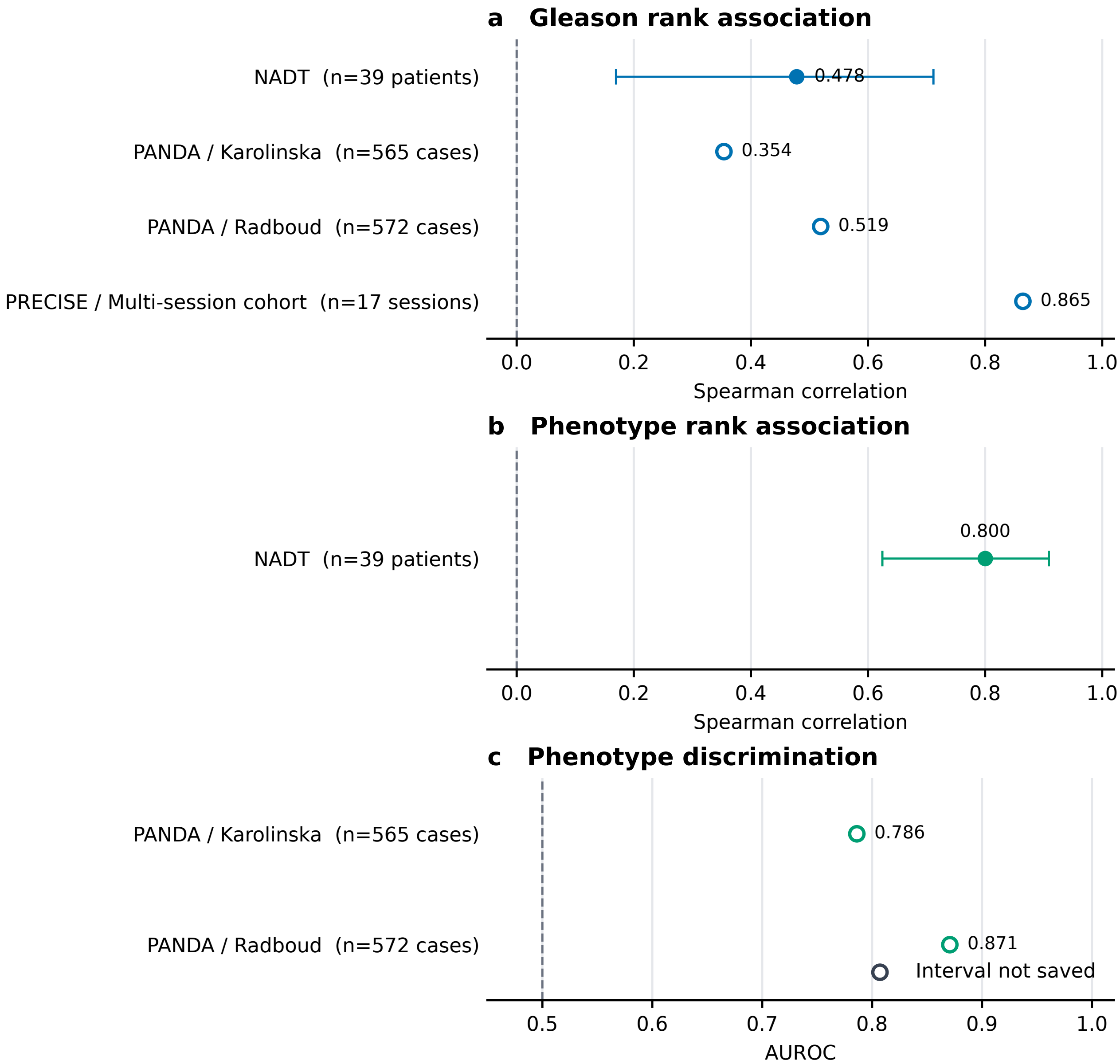


Figure S1: **Saved grade and phenotype transport details.** Aggregate patient-, case-image-, or session-level estimates are separated by metric: **a**, Gleason Spearman correlations in NADT-Prostate, PANDA, and PRECISE; **b**, NADT-Prostate phenotype Spearman correlation; and **c**, phenotype AUROC after transfer without refitting to the Karolinska and Radboud PANDA subsets. Filled points with whiskers show the saved estimate and patient-bootstrap interval; open points denote estimates for which no interval was saved. Dashed lines mark 0 for correlation and 0.5 for AUROC. Denominators and analysis units are printed for every row. Unavailable intervals were not treated as zero, and aggregate statistics were not converted into reconstructed curves.

Recurrence endpoints, paired bootstrap and common-source sensitivity

This subsection tests how the recurrence interpretation changes with endpoint definition, paired uncertainty estimation, and restriction to a common patient source frame.

The recurrence endpoints were analysed separately. On the 270-patient target frame, official TCGA-CDR PFI had 42 events and the reconstructed with-tumor endpoint had 57. Their event agreement was 0.870, Cohen's $\kappa = 0.569$, and follow-up-time Spearman $\rho = 0.849$. Official PFI and the cBioPortal TCGA-CDR PFS fields agreed exactly on the 270 common evaluable patients; this identity does not make the separately reconstructed endpoint equivalent to PFI. Among 192 patients shared with DFS, event agreement was 0.979 and $\kappa = 0.835$. Among 220 patients shared with the strict recurrence-only sensitivity, event agreement was 0.955 and $\kappa = 0.564$. These endpoint concordance results come from `models/pfi_endpoint_concordance.csv`. Cohen's kappa measures agreement after accounting for agreement expected by chance.

The paired survival analysis used 153 complete cases for each endpoint, with 30 reconstructed-endpoint events and 15 official-PFI events. Every reported model and contrast requested 2,000 patient-bootstrap resamples. The saved accounting was 2,000 valid and zero undefined bootstrap replicates (undefined fraction 0) for all 44 model-metric summaries and all 20 paired contrast-metric summaries. Zero undefined replicates is reported rather than silently assumed; another dataset could yield undefined concordance or Brier-score draws.

The post-hoc recurrence-risk source-retention audit compared all 60 seed cells on the saved source sets with the same cells restricted to the common-498 patient intersection. No cell crossed the C-index null, and all 65 paired contrast directions were preserved. The largest absolute cell shift was 0.023. This common-498 sensitivity limits source-retention confounding in the saved grid but does not provide an independent cohort or a causal representation comparison.

Supplementary Figure S2 displays the saved time-dependent AUROC and calibration summaries for the reconstructed with-tumor endpoint only. This legacy figure source is not Official TCGA-CDR PFI. Its endpoint identity and analysis frame were accepted only after the calibration quintiles at both horizons reconciled to the saved frozen-risk metadata row (270 patients and 57 events). The exploratory marker is not interpreted as an independently validated prognostic biomarker.

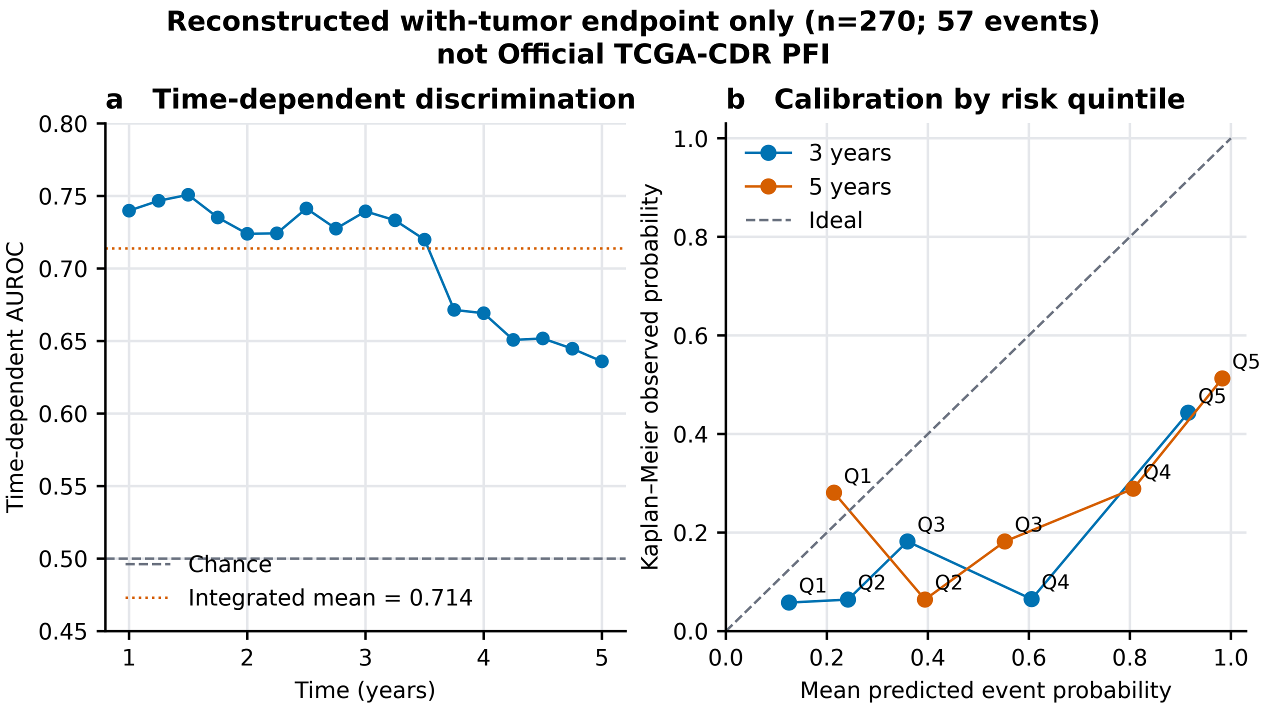


Figure S2: **Legacy reconstructed-endpoint post-hoc recurrence-risk summaries.** Values are rendered from saved legacy sources for the Reconstructed with-tumor endpoint (270 patients; 57 events), not Official TCGA-CDR PFI. **a**, Time-dependent AUROC; the dashed line is chance (0.5), and the dotted line is the saved integrated mean over 1–5 years. **b**, Mean predicted event probability versus Kaplan–Meier observed probability by risk quintile (Q1–Q5) at 3 and 5 years; the diagonal is ideal calibration and connecting lines are visual guides across quintiles. The endpoint identity, denominator, event count, and horizon-specific calibration totals were reconciled to the saved frozen-risk metadata. Kaplan–Meier is a nonparametric estimate of observed event probability; calibration compares predicted and observed probabilities; and risk quintiles divide patients into five ordered groups of predicted risk. This legacy summary is exploratory and supplies no Official-PFI calibration claim.

Full correlated stability analysis

We next expose the complete sensitivity grid behind the condensed stability decision in the main manuscript while preserving its correlated, non-confirmatory interpretation.

The saved grid has 72 configurations: six targets, two encoders, two physical scales per encoder and three tile budgets. Each configuration contains five repeated sampling seeds, giving 360 correlated seed cells. All settings reused the underlying cohorts and folds. The five-seed Student-t intervals therefore describe tile-sampling variation, not patient or population uncertainty. The saved paired-contrast table contains 390 seed-matched comparisons: 180 native-versus-shared-scale, 90 Virchow-versus-CONCH at shared scale and 120 64-versus-16-tile comparisons.

Supplementary Table S3 separates the observed seed range from the descriptive Student-t interval. Supplementary Figure S3 contains the complete configuration means followed by the paired contrasts. A setting-specific direction does not identify a causal effect of encoder, scale or tile count, and the 360 cells are not independent validations.

Table S3: Global seed-cell ranges and configuration-level null straddling recomputed from all 72 saved configurations. The last two columns count 12 configurations per target and distinguish the observed five-seed range from the descriptive Student-t interval.

Target	Global minimum	Global maximum	Range straddles	t-interval contains null
Gleason	0.271	0.646	0/12	0/12
Phenotype	0.845	0.965	0/12	0/12
PTEN	0.519	0.717	0/12	0/12
AR	0.136	0.297	0/12	0/12
SPOP	0.348	0.679	8/12	9/12
Recurrence risk	0.481	0.755	1/12	2/12

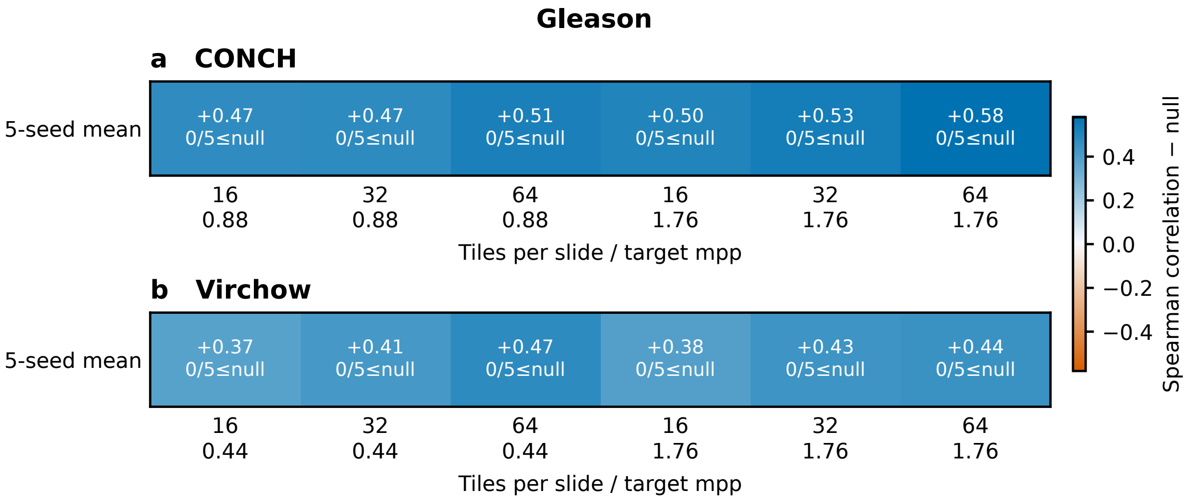
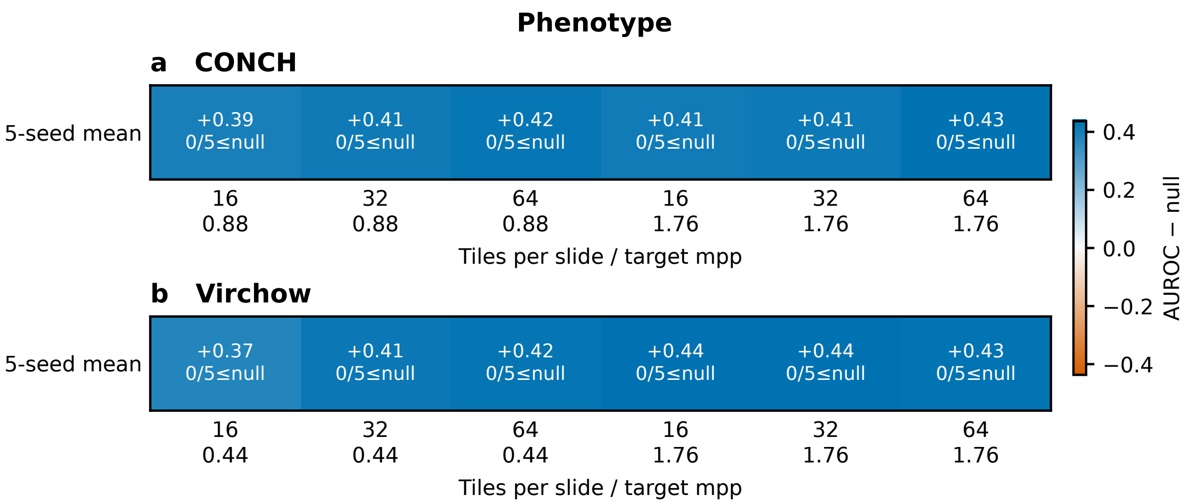
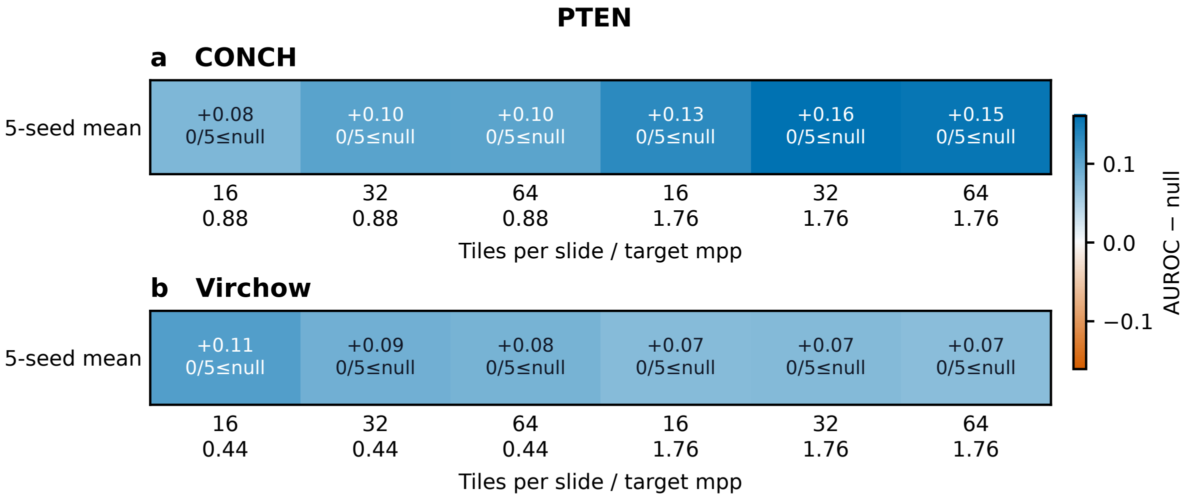


Figure S3: **Complete stability grids and paired contrasts.** Pages 1–6 display all 72 validated configuration means from the saved grid, one target per page. On each page, panels **a–b** are CONCH and Virchow; columns combine 16, 32, or 64 tiles with the encoder-native physical scale or the shared 1.76 μm-per-pixel scale. Each cell reports the five-seed mean minus the marker-specific null and the number of seeds at or below that null; color is centred at zero separately for each target. Pages 7–9 display all 390 seed-matched contrasts: shared 1.76 μm-per-pixel minus native scale (180), Virchow minus CONCH at 1.76 μm per pixel (90), and 64 minus 16 tiles (120). The abscissa is metric $B - A$ and the dashed line is zero; orange points cross the marker-specific null whereas blue points remain on the same side. These are descriptive setting checks on reused cohorts and folds, not independent validations or population-confidence intervals.

Supplementary Figure S3 (continued; page 2 of 9)



Supplementary Figure S3 (continued; page 3 of 9)



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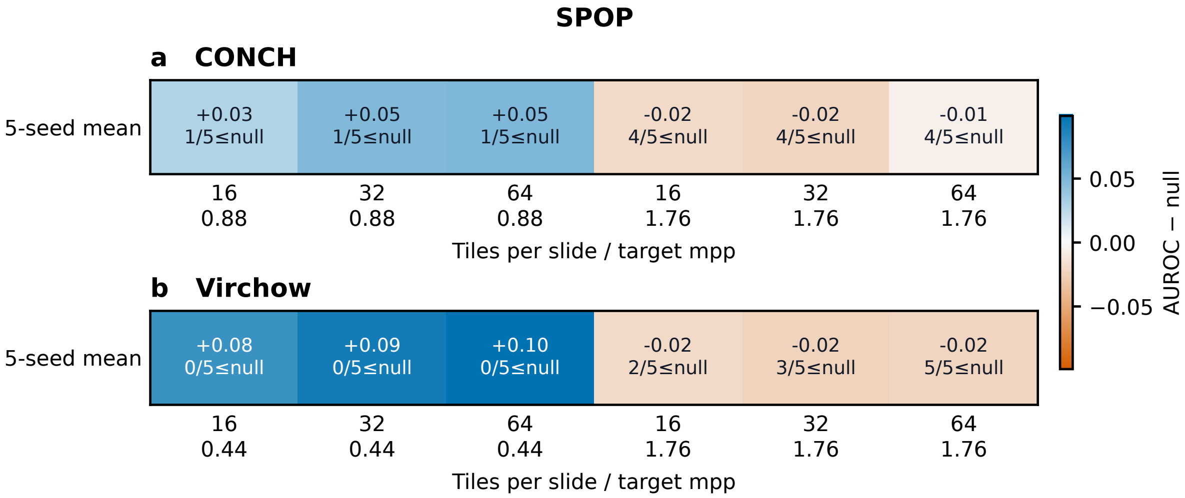
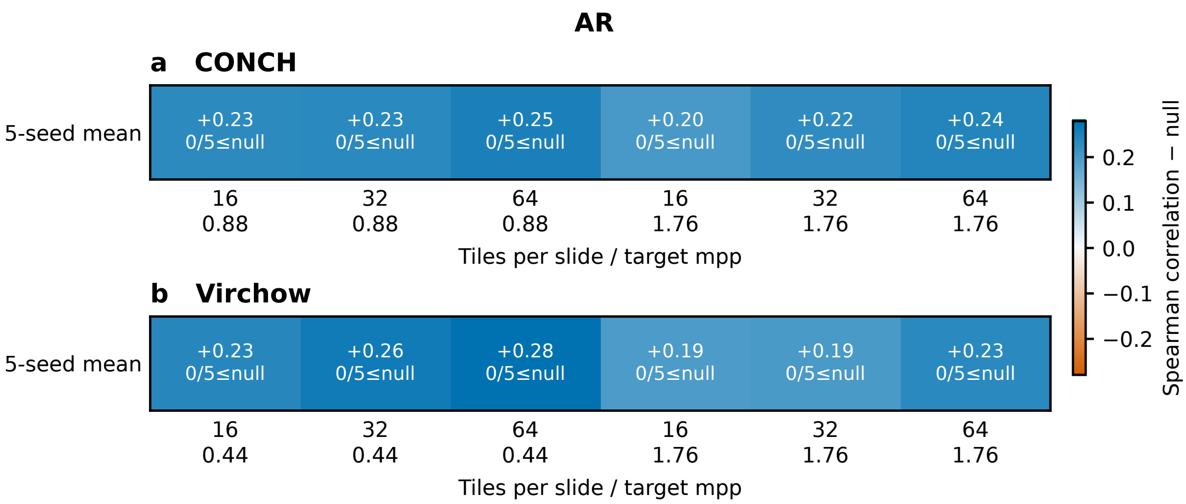


Supplementary Figure S3 (continued; page 5 of 9)



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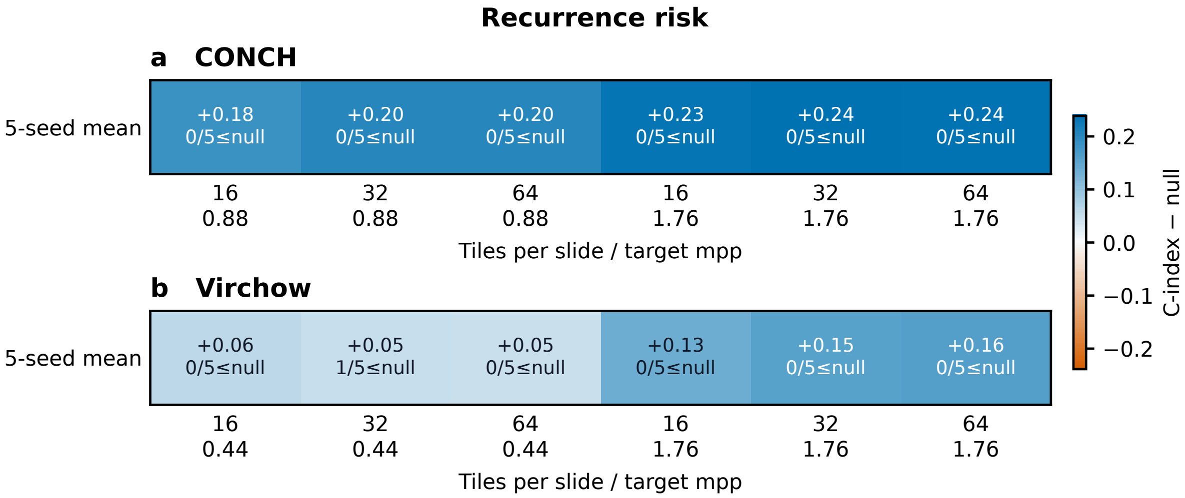

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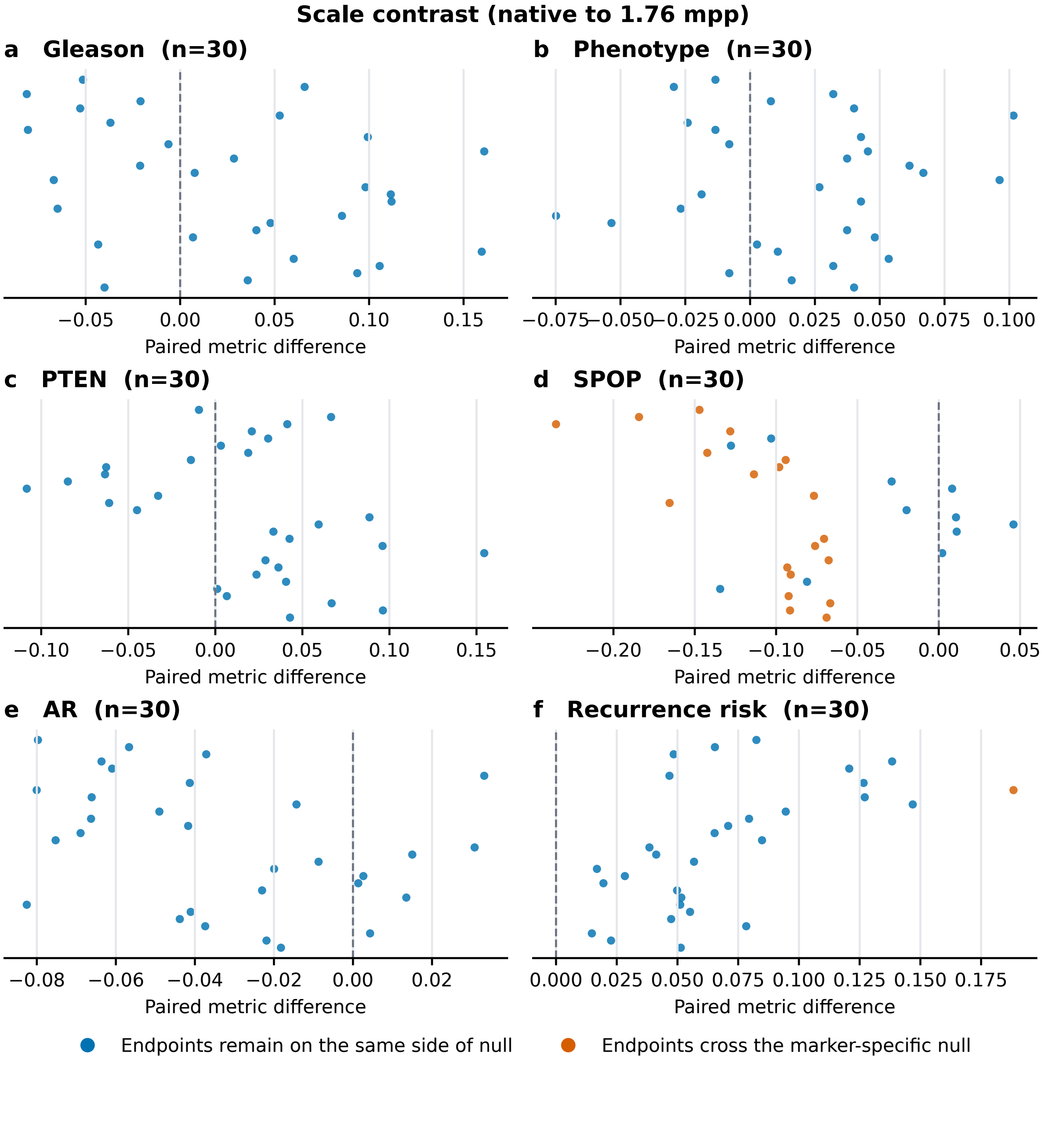

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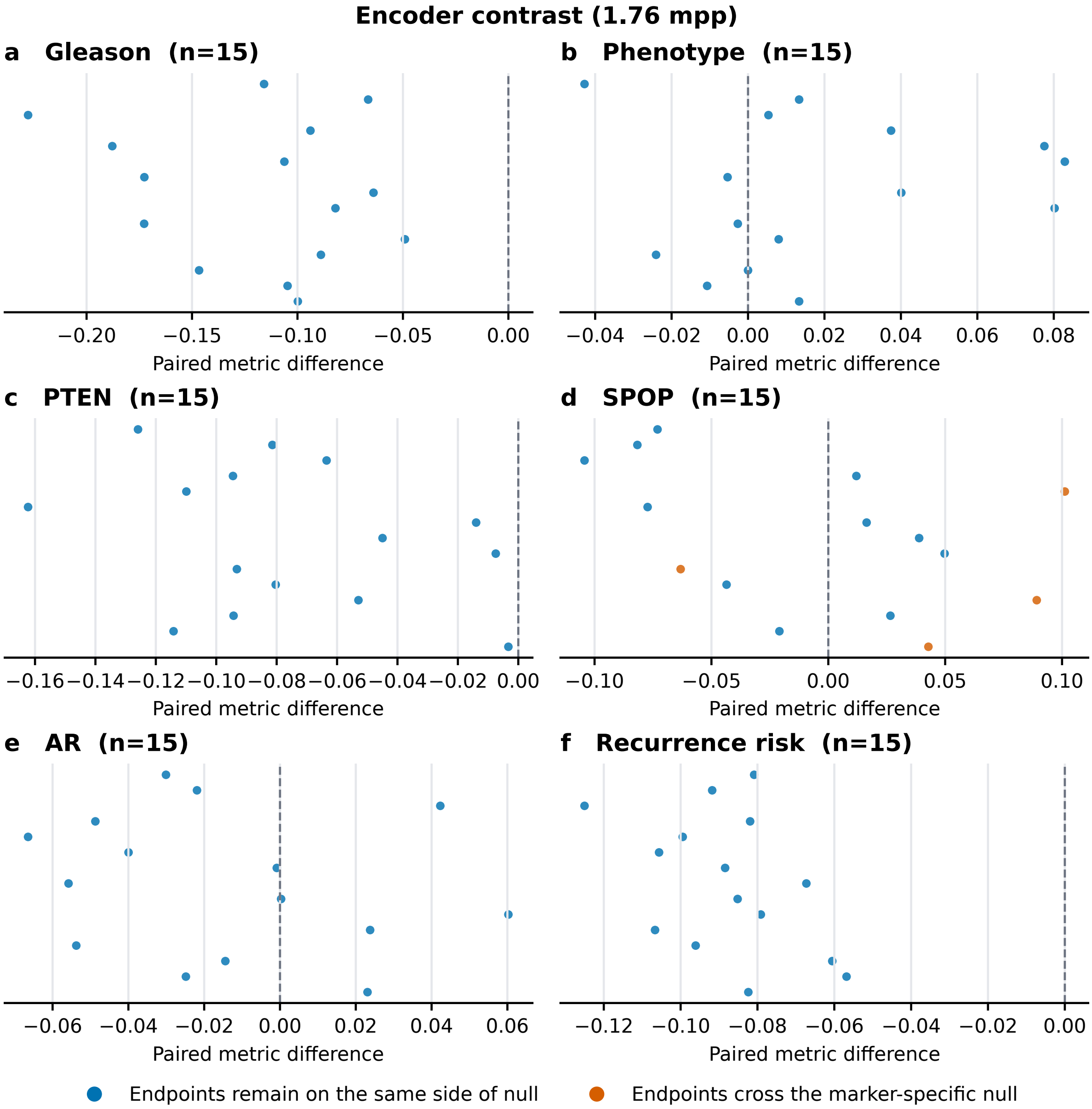
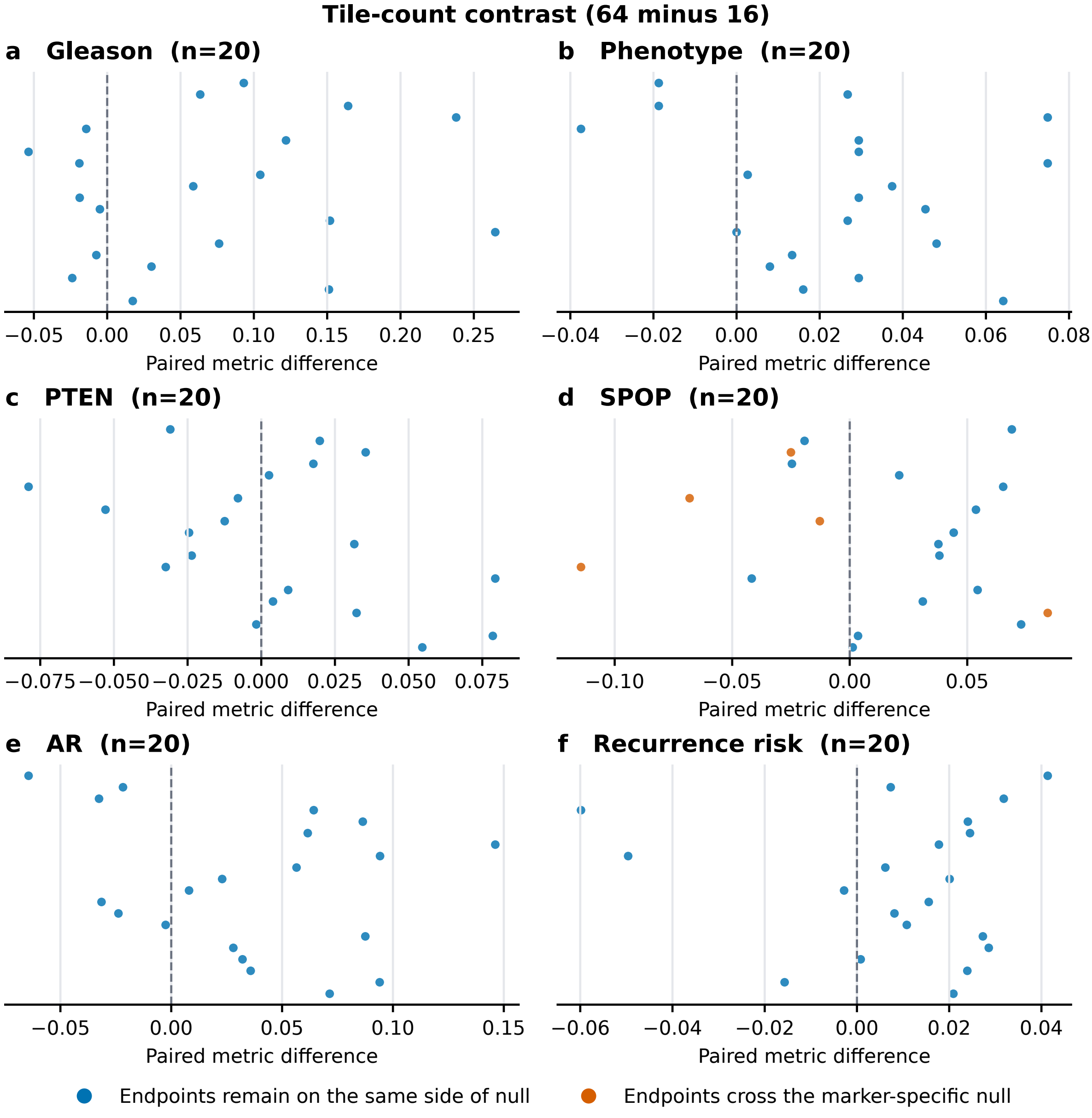


Supplementary Figure S3 (continued; page 9 of 9)



Target-by-axis evidence coverage

This subsection makes the synthesis explicit by showing which evidence axis supports, restricts, or remains unavailable for each target.

The qualification framework deliberately preserves both positive evidence and evidentiary gaps. Supplementary Figure S4 shows the complete matrix of frozen-primary support, cross-cohort transport, clinical increment, site behavior, setting stability and endpoint fidelity for each target group. A supported cell refers only to the scope stated for that axis; it does not fill an unevaluated cell elsewhere. Supplementary Table S4 links every cell to the audited claim source, states the current interpretation and identifies the evidence required to change that state.

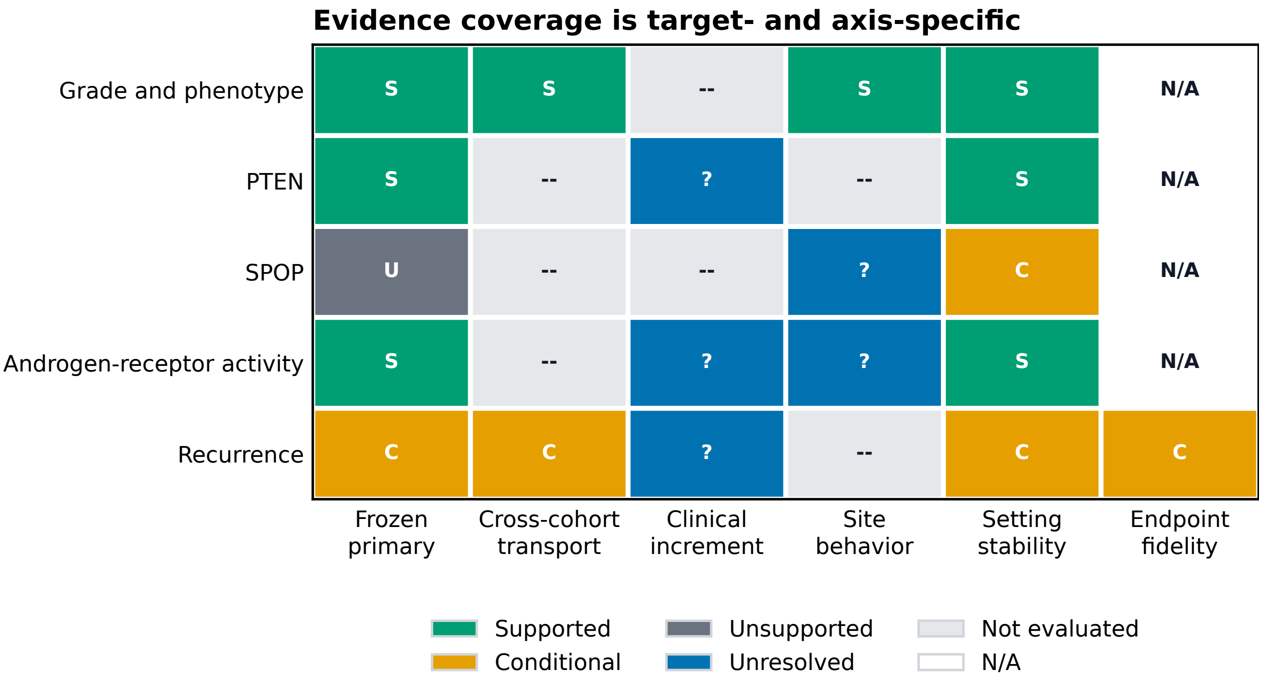


Figure S4: **Target-by-axis evidence coverage.** Each cell records the evidence state supported by the frozen analyses for one target row and one evidentiary-axis column. Columns separate frozen-primary support, cross-cohort transport, clinical increment, site behavior, setting stability, and endpoint fidelity. Cell labels are S (supported), C (conditional), U (unsupported in the frozen primary design), ? (evaluated but unresolved), – (not evaluated), and N/A (not applicable). A supported or conditional cell is bounded to its named axis and cannot fill another cell; not-evaluated and not-applicable cells are not negative evidence. The source-linked audit table below provides the claim basis and evidence needed to change every cell.

Table S4: Target-by-axis qualification audit. Each row links an evidence state to the audited claim source and states what evidence would be required to change that state.

Target	Evidence axis	State	Claim	Current evidence	Next evidence needed
Grade and phenotype	Frozen primary	supported	C02	Source-cohort grade and phenotype signals were supported.	Retain the locked source probes and analysis units.
Grade and phenotype	Cross-cohort transport	supported	C02	Direction was retained without refitting in PANDA and was concordant with PRECISE grade evaluation.	Add patient-level uncertainty where source data permit and test further institutions.
Grade and phenotype	Clinical increment	not evaluated	C02	No incremental clinical-covariate claim was defined for these morphology-linked targets.	Define a clinically meaningful reference model before making an increment claim.
Grade and phenotype	Site behavior	supported	C02	Transfer directions were retained at both PANDA institutions; this was not a causal site analysis.	Use site-specific patient-level intervals and acquisition metadata in additional cohorts.
Grade and phenotype	Setting stability	supported	C02;C08	Correlated setting audits retained the transported directions.	Confirm stability on independently sampled patients rather than reused settings.
Grade and phenotype	Endpoint fidelity	not applicable	C02	No time-to-event endpoint was used for these targets.	Not applicable to the present grade and phenotype claims.
PTEN	Frozen primary	supported	C03	The within-TCGA frozen-primary association was above chance.	Repeat the locked evaluation in an external cohort with compatible PTEN labels.
PTEN	Cross-cohort transport	not evaluated	C03	No compatible external PTEN cohort was available in this study.	Acquire an external cohort with target-matched PTEN assays.
PTEN	Clinical increment	unresolved	C03	Held-out increment beyond grade was not established.	Test a locked image increment against grade and fuller clinical covariates in new patients.
PTEN	Site behavior	not evaluated	C03	The manuscript does not establish external site transport for PTEN.	Use a multisite external design with site and acquisition metadata.
PTEN	Setting stability	supported	C03;C08	The within-cohort direction remained above chance across the evaluated grid.	Confirm the direction under independent cohort resampling.
PTEN	Endpoint fidelity	not applicable	C03	PTEN loss was a binary molecular target rather than a clinical time-to-event endpoint.	Not applicable to the present PTEN claim.
SPOP	Frozen primary	unsupported in frozen design	C04	The frozen primary configuration did not support the proposed effect.	Increase target-appropriate sampling without changing the frozen primary interpretation.
SPOP	Cross-cohort transport	not evaluated	C04	No compatible external SPOP cohort was evaluated.	Test a locked probe in an external cohort with mutation labels and tumor-content control.
SPOP	Clinical increment	not evaluated	C04	A grade-independent SPOP increment was not established in the approved analysis set.	Evaluate incremental value with patient-disjoint nested analysis if scientifically justified.
SPOP	Site behavior	unresolved	C04	Defined site intervals included or touched chance and one site was single-class.	Collect larger site-balanced data with assay and acquisition metadata.
SPOP	Setting stability	context-sensitive	C04;C08	The result crossed the null across multiple correlated settings.	Resolve tumor sampling and representation sensitivity in independent data.

Target	Evidence axis	State	Claim	Current evidence	Next evidence needed
SPOP	Endpoint fidelity	not applicable	C04	SPOP mutation was a binary molecular target rather than a clinical time-to-event endpoint.	Not applicable to the present SPOP claim.
Androgen-receptor activity	Frozen primary	supported	C05	The pooled frozen-primary direction was positive.	Retain the pooled result as within-cohort evidence only.
Androgen-receptor activity	Cross-cohort transport	not evaluated	C05	No compatible external androgen-receptor activity cohort was evaluated.	Use an external cohort with a compatible activity assay.
Androgen-receptor activity	Clinical increment	unresolved	C05	Held-out grade-independent increment was not established.	Test a locked incremental model against grade and fuller clinical structure.
Androgen-receptor activity	Site behavior	unresolved	C05	Every stored site-specific interval crossed zero and acquisition effects could not be separated.	Disentangle site from scanner stain preparation and case mix.
Androgen-receptor activity	Setting stability	supported	C05;C08	The positive direction was retained across the correlated setting grid.	Confirm with independently sampled sites and patients.
Androgen-receptor activity	Endpoint fidelity	not applicable	C05	Androgen-receptor activity was a continuous molecular target rather than a time-to-event endpoint.	Not applicable to the present activity-score claim.
Recurrence	Frozen primary	context-sensitive	C06	The transferred post-hoc risk signal depended on representation and endpoint.	Lock the source model before a new independent target-cohort evaluation.
Recurrence	Cross-cohort transport	context-sensitive	C06	LEOPARD-to-TCGA transfer was present in some settings but was not endpoint-invariant.	Test the locked score in an external cohort with a harmonized endpoint.
Recurrence	Clinical increment	unresolved	C07	Full clinical and site comparisons did not establish robust incremental value.	Use an adequately powered same-patient comparison against a clinical reference locked before evaluation.
Recurrence	Site behavior	not evaluated	C06;C07	Site was included in fuller models but site-specific recurrence transport was not isolated.	Evaluate site transport with sufficient events per site.
Recurrence	Setting stability	context-sensitive	C06;C08	Effect size and direction depended on encoder scale and related settings.	Use nested locked model selection and independent validation.
Recurrence	Endpoint fidelity	context-sensitive	C06;C07	Reconstructed recurrence and official PFI produced different evidentiary conclusions.	Harmonize endpoint definitions and retain endpoint-specific reporting.

AR site uncertainty and SPOP detectable-effect calculations

We then provide the detailed checks that bound the site-specific AR interpretation and the effect sizes that the frozen SPOP analysis was positioned to detect.

The AR site audit estimated slide-level Spearman correlations and used patient-cluster bootstrap intervals. Six leave-one-site-out rows were available: CH, −0.128 (−0.561 to 0.350); EJ, 0.127 (−0.121 to 0.363); G9, 0.195 (−0.258 to 0.598); HC, 0.217 (−0.072 to 0.471); KK, 0.265 (−0.087 to 0.558); and YL, 0.227 (−0.135 to 0.541). Every site-specific interval included zero. The positive pooled patient-level estimate therefore did not establish transport at any individual site, and the data did not identify scanner or stain as a cause.

For SPOP, the saved planning calculation used 29 positive and 244 negative patients, a two-sided $\alpha = 0.05$ test and a fixed-score Hanley–McNeil normal approximation. The minimum detectable AUROC was 0.661 at 80% power and 0.685 at 90% power. These values omit probe-training uncertainty, out-of-fold dependence, site heterogeneity, setting selection and multiplicity. They are planning approximations, not effect-exclusion bounds; they do not show that smaller effects or sampling dilution are absent. In this calculation, "minimum detectable" means the smallest assumed AUROC that reaches the specified power under the stated assumptions, not the smallest biologically possible effect.

Provenance and reproducibility scope

The final subsection identifies the saved records supporting computational traceability and distinguishes that traceability from external or clinical validation.

The extended results were read from saved CSV sources rather than recomputed from remembered values. Stability lineage is recorded in `models/stability_run_manifest.csv` and `models/stability_qc_report.json`; recurrence lineage is recorded in the paired survival and common-source run configurations and manifests; AR and SPOP source hashes are recorded in `models/ar_spop_evidence_manifest.csv`. The manuscript source tables and figure renderers are linked by the submission figure manifest. The PRECISE clinician review source was kept immutable and its SHA256 was checked against `c1dd522b4ff4f233b3a23630bf9074da881bb7b9145996fc47c3383a0448d2a3`. A SHA256 hash is a file fingerprint used to detect byte-level changes, while a manifest is an inventory linking named inputs, code, configurations and outputs.

These records support auditability of the saved analyses. They do not establish population prevalence, whole-slide diagnostic performance, external molecular validation or clinical utility.